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Final Project

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1 INTRODUCTION

This project analyzes a single-stage vapor-compression refrigeration system for data center cooling. The assigned system has a nominal cooling capacity of 50 kW, with air as the secondary fluid on the evaporator side and water as the secondary fluid on the condenser side.

The main objective is to evaluate how the refrigerating capacity Q_e and the Energy Efficiency Ratio (EER) vary when the condenser heat-sink temperature changes. For this purpose, the condenser water inlet temperature is varied from 20°C to 40°C, while the other main operating assumptions are kept fixed.

The system is modeled using R32 as the refrigerant and a Bitzer scroll compressor. Compressor performance is evaluated using manufacturer polynomial data, while refrigerant thermodynamic properties are calculated with CoolProp. The evaporator and condenser are modeled with the LMTD method, and the evaporating and condensing temperatures are obtained through an iterative coupling between the refrigeration cycle and the heat exchangers.

Table 1: Main project specifications

Quantity	Value
Application	Data center cooling
Nominal cooling capacity	50 kW
Assigned evaporator air condition	12°C
Condenser secondary fluid	Water
Refrigerant	R32
Selected compressor	Bitzer GSU60182VL.4
Variable parameter	Condenser water inlet temperature
Water inlet temperature range	20°C–40°C

2 DESIGN CHOICES AND MAIN ASSUMPTIONS

2.1 FROM SECONDARY-FLUID TEMPERATURES TO REFRIGERANT-SIDE TEMPERATURES

The assignment defines the application and the secondary-fluid conditions, but the compressor selection requires refrigerant-side saturation temperatures. In particular, the Bitzer software requires the evaporating saturation temperature T_{evap} and the condensing saturation temperature T_{cond} .

These refrigerant-side temperatures are not equal to the secondary-fluid temperatures because heat transfer in the evaporator and condenser requires a finite temperature difference. Therefore, approach-temperature assumptions are used to estimate a reasonable nominal compressor selection point.

For a heat exchanger, the refrigerant-side temperature is estimated from the average secondary-fluid temperature and a typical approach temperature. On the condenser side, where heat is rejected to water,

$$T_{\text{cond}} = \bar{T}_{\text{w,c}} + \Delta T_{\text{cond,app}}, \quad (1)$$

where $\bar{T}_{\text{w,c}}$ is the average condenser water temperature and $\Delta T_{\text{cond,app}}$ is the condenser approach temperature. On the evaporator side, where heat is absorbed from air,

$$T_{\text{evap}} = T_{\text{air,out}} - \Delta T_{\text{evap,app}}. \quad (2)$$

For the condenser, the nominal water inlet temperature is taken as 30°C, and the water temperature rise is fixed at 5 K. Therefore,

$$T_{\text{w,in}} = 30^\circ\text{C}, \quad T_{\text{w,out}} = 35^\circ\text{C}. \quad (3)$$

The average condenser water temperature is then

$$\bar{T}_{\text{w,c}} = \frac{T_{\text{w,in}} + T_{\text{w,out}}}{2} = \frac{30 + 35}{2} = 32.5^\circ\text{C}. \quad (4)$$

Using a typical water-side condenser approach temperature of 5 K, the nominal condensing temperature becomes

$$T_{\text{cond}} = 32.5 + 5 = 37.5^{\circ}\text{C}. \quad (5)$$

For the evaporator, the assignment specifies an air condition of 12°C . In this model, this value is interpreted as the cold supply air temperature delivered to the data center. To define the air-side cooling process, the return air temperature is assumed to be 24°C . Thus,

$$T_{\text{air,in}} = 24^{\circ}\text{C}, \quad T_{\text{air,out}} = 12^{\circ}\text{C}. \quad (6)$$

Since the evaporator absorbs heat from air, a typical air-side evaporator approach temperature of 10 K is used. Therefore,

$$T_{\text{evap}} = T_{\text{air,out}} - 10 = 12 - 10 = 2^{\circ}\text{C}. \quad (7)$$

The nominal refrigerant-side temperatures used for compressor selection are therefore

$$T_{\text{evap}} = 2^{\circ}\text{C}, \quad T_{\text{cond}} = 37.5^{\circ}\text{C} \quad (8)$$

These values are used only to select and validate the compressor at a nominal operating point. In the final system model, T_{evap} and T_{cond} are not imposed during the parametric study. Instead, they are calculated iteratively from the compressor model and the heat exchanger balances.

2.2 COMPRESSOR TYPE SELECTION

A scroll compressor was selected because the assigned cooling capacity is in the small-to-medium HVAC range, where scroll compressors are commonly suitable. For a data center cooling application, the compressor should be compact, reliable, and efficient over the expected range of evaporating and condensing temperatures.

Compared with other compressor types, a scroll compressor offers a good compromise between efficiency, simplicity, low vibration, and compactness. Reciprocating compressors are also possible at this capacity range, but they generally have more moving parts and higher mechanical complexity. Screw compressors are typically more suitable for larger cooling capacities, while centrifugal compressors are not appropriate for this scale.

Table 2: Qualitative comparison of compressor types for the assigned application

Compressor type	Suitability for this project
Scroll	Suitable for small-to-medium HVAC capacity; compact and efficient
Reciprocating	Possible, but mechanically more complex
Screw	More suitable for larger cooling capacities
Centrifugal	Not appropriate for this capacity range

2.3 COMPRESSOR MODEL SELECTION

Using the nominal refrigerant-side temperatures derived above, two close Bitzer R32 scroll compressor options were compared: GSU60182VL.4 and GSD60235VL.4. The comparison was performed at

$$T_{\text{evap}} = 2^{\circ}\text{C}, \quad T_{\text{cond}} = 37.5^{\circ}\text{C}, \quad (9)$$

with 6 K suction superheating and 3 K liquid subcooling.

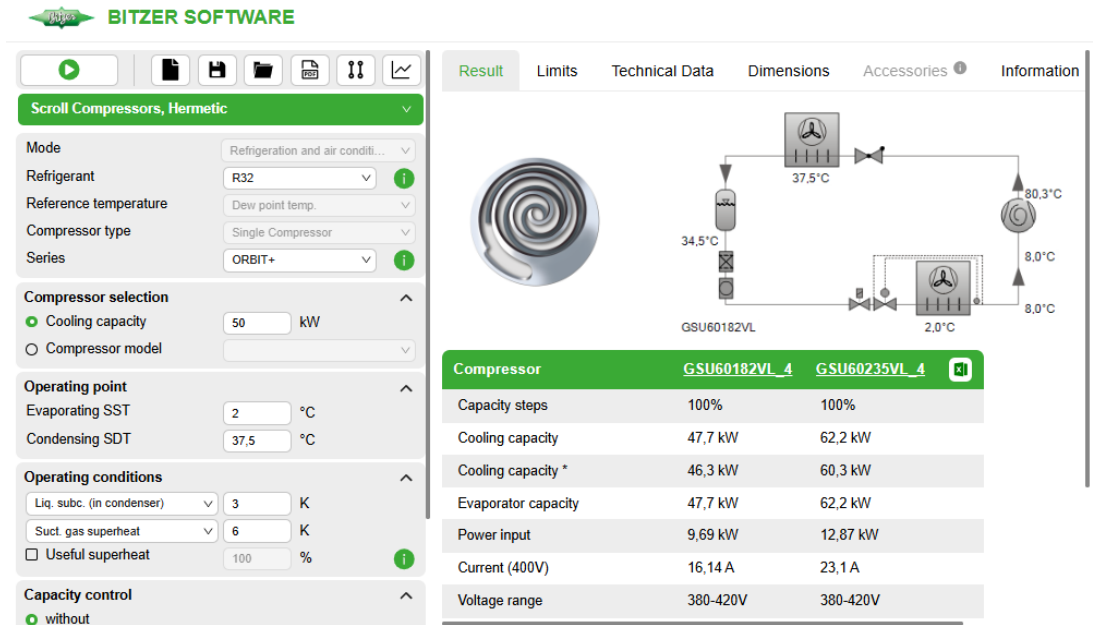


Figure 1: Bitzer compressor selection at the nominal operating point for R32, with $T_{\text{evap}} = 2^\circ\text{C}$, $T_{\text{cond}} = 37.5^\circ\text{C}$, 6 K suction superheating, and 3 K liquid subcooling.

The selected compressor is the Bitzer GSU60182VL_4. At the nominal point, it provides a cooling capacity of 47.7 kW, which is slightly below the assigned 50 kW target. The deviation is

$$\frac{47.7 - 50}{50} \times 100 = -4.6\%. \quad (10)$$

This deviation is considered acceptable because commercial compressors are available only in discrete sizes. The alternative compressor, GSD60235VL_4, provides 60.2 kW, corresponding to

$$\frac{60.2 - 50}{50} \times 100 = +20.4\%. \quad (11)$$

Therefore, although the selected compressor is slightly undersized, it is much closer to the nominal assignment capacity than the larger alternative.

Table 3: Comparison of candidate Bitzer R32 scroll compressors at the nominal operating point

Compressor	Q_e [kW]	Deviation	P_c [kW]	COP/EER	Decision
GSU60182VL_4	47.7	-4.6%	9.69	4.92	Selected
GSD60235VL_4	60.2	+20.4%	13.27	4.54	Rejected

Values are taken from the Bitzer selection software at $T_{\text{evap}} = 2^\circ\text{C}$, $T_{\text{cond}} = 37.5^\circ\text{C}$, 6 K suction superheating, and 3 K liquid subcooling.

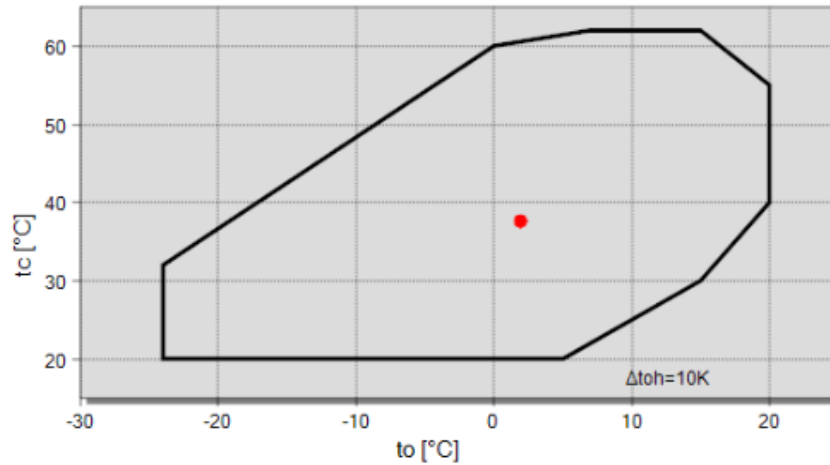


Figure 2: Operating envelope of the selected Bitzer GSU60182VL_4 compressor. The nominal operating point lies inside the permissible operating region.

The red point in Figure 2 corresponds to the nominal compressor-selection condition,

$$T_{\text{evap}} = 2^{\circ}\text{C}, \quad T_{\text{cond}} = 37.5^{\circ}\text{C}.$$

Since this point lies inside the permissible operating envelope, the selected evaporating and condensing temperatures are compatible with the compressor operating limits. The selected compressor data are summarized in Table 4.

Table 4: Selected compressor data

Quantity	Value
Manufacturer	Bitzer
Model	GSU60182VL_4
Series	ORBIT+
Type	Scroll compressor
Refrigerant	R32
Cooling capacity at nominal point	47.7 kW
Compressor power at nominal point	9.69 kW
Mass flow rate at nominal point	664 kg/h
COP/EER at nominal point	4.92
Swept volume at 50 Hz	30.2 m ³ /h
Maximum power input	16.7 kW
Maximum pressure LP/HP	34.2/45 bar

2.4 REFRIGERANT SELECTION

The selected refrigerant is R32. The choice was constrained by the refrigerants available in the Bitzer software for the selected scroll compressor family and by the need for a suitable refrigerant for data center cooling.

The comparison considers three main criteria: environmental impact, safety classification, and practical availability of compressor performance data. The ozone depletion potential (ODP), global warming potential (GWP), and safety classification values are reported according to standard refrigerant classification and environmental reference data. The compressor compatibility and performance data are taken from the Bitzer software.

R32 was selected as the best practical compromise because it provides good performance, is supported by manufacturer compressor data, has zero ozone depletion potential, and has lower GWP than older refrigerants such as R410A, R134a, R404A, and R507A. Its A2L safety class means that it is mildly flammable, so safety requirements

must be considered in real installations. However, for the scope of this thermodynamic modeling project, R32 is a suitable choice because it combines performance, availability, and lower environmental impact compared with many conventional HFC refrigerants.

Table 5: Qualitative comparison of refrigerants considered in the Bitzer selection. ODP, GWP, and safety class values are based on standard refrigerant classification and environmental reference data.

Refrigerant	ODP	Approx. GWP	Safety	Comment
R32	0	~ 675	A2L	Selected; good performance and available Bitzer data
R454B	0	~ 466	A2L	Lower GWP; strong alternative
R452B	0	~ 676	A2L	Similar replacement option for R410A
R290	0	~ 3	A3	Very low GWP, but stricter flammability constraints
R410A	0	~ 2088	A1	Good performance, but high GWP
R134a	0	~ 1430	A1	Non-flammable, but high GWP
R404A/R507A	0	Very high	A1	Avoided due to very high GWP
R22	Nonzero	High	A1	Avoided due to ozone depletion

2.5 SUMMARY OF MAIN OPERATING ASSUMPTIONS

The main operating assumptions used in the model are summarized in Table 6. The suction superheating and liquid subcooling values are the same values used in the Bitzer compressor selection. Superheating ensures dry vapor at the compressor inlet, while subcooling ensures liquid refrigerant at the expansion valve inlet.

Table 6: Main operating assumptions used in the model

Assumption	Value	Reason
Evaporator air inlet	24°C	Assumed data center return air temperature
Evaporator air outlet	12°C	Assigned air condition, treated as supply air
Nominal evaporating temperature	2°C	10 K air-side approach below supply air
Nominal condensing temperature	37.5°C	5 K approach above average condenser water temperature
Suction superheating	6 K	Dry vapor at compressor inlet; Bitzer setting
Liquid subcooling	3 K	Liquid at expansion valve inlet; Bitzer setting
Condenser water rise	5 K	Fixed heat-sink temperature rise
Pressure drops	Neglected	Simplifying assumption for system-level model

3 THERMODYNAMIC AND COMPRESSOR MODEL

3.1 VAPOR-COMPRESSION CYCLE

The system is modeled as a single-stage vapor-compression refrigeration cycle. The four main thermodynamic states are:

- State 1: compressor inlet, superheated vapor;
- State 2: compressor outlet, superheated high-pressure vapor;
- State 3: condenser outlet, subcooled liquid;
- State 4: expansion valve outlet, two-phase mixture.

The main processes are compression, heat rejection in the condenser, throttling through the expansion valve, and heat absorption in the evaporator. The refrigerant properties at each state are calculated using CoolProp.

The compressor inlet temperature is defined from the evaporating temperature and the imposed suction superheating:

$$T_1 = T_{\text{evap}} + \Delta T_{\text{sh}}. \quad (12)$$

The condenser outlet temperature is defined from the condensing temperature and the imposed liquid subcooling:

$$T_3 = T_{\text{cond}} - \Delta T_{\text{sc}}. \quad (13)$$

The expansion valve is modeled as an isenthalpic process:

$$h_4 = h_3. \quad (14)$$

The compressor outlet enthalpy is reconstructed from the compressor power and refrigerant mass flow rate:

$$h_2 = h_1 + \frac{P_c}{\dot{m}}. \quad (15)$$

3.2 COMPRESSOR POLYNOMIAL MODEL

The compressor is modeled using manufacturer polynomial data exported from the Bitzer software. This approach is more realistic than assuming constant compressor efficiencies because it accounts for the dependence of compressor performance on evaporating and condensing temperatures.

For each operating point, the polynomial model provides:

$$Q_{e,\text{poly}}, \quad P_{c,\text{poly}}, \quad \dot{m}. \quad (16)$$

The same 10-coefficient polynomial form is used for cooling capacity, compressor power, and refrigerant mass flow rate:

$$y = c_1 + c_2 T_{\text{evap}} + c_3 T_{\text{cond}} + c_4 T_{\text{evap}}^2 + c_5 T_{\text{evap}} T_{\text{cond}} + c_6 T_{\text{cond}}^2 + c_7 T_{\text{evap}}^3 + c_8 T_{\text{cond}} T_{\text{evap}}^2 + c_9 T_{\text{evap}} T_{\text{cond}}^2 + c_{10} T_{\text{cond}}^3, \quad (17)$$

where

$$y \in \{Q_e, P_c, \dot{m}\}. \quad (18)$$

The temperatures in the polynomial are expressed in degrees Celsius. The mass flow rate from the Bitzer polynomial is given in kg/h and is converted to kg/s in the model.

3.3 CYCLE PERFORMANCE CALCULATION

After obtaining P_c and $\dot{m}$ from the compressor polynomial, the thermodynamic cycle is reconstructed using refrigerant properties from CoolProp.

The evaporator cooling capacity is calculated as:

$$Q_e = \dot{m} (h_1 - h_4). \quad (19)$$

The compressor power is:

$$P_c = \dot{m} (h_2 - h_1). \quad (20)$$

The condenser heat rejection is:

$$Q_c = \dot{m} (h_2 - h_3). \quad (21)$$

The Energy Efficiency Ratio is defined as:

$$\text{EER} = \frac{Q_e}{P_c}. \quad (22)$$

The pressure ratio is calculated from the condensing and evaporating pressures:

$$PR = \frac{P_{\text{cond}}}{P_{\text{evap}}}. \quad (23)$$

3.4 DERIVED COMPRESSOR EFFICIENCIES

The polynomial model directly gives P_c and $\dot{m}$, but the isentropic and volumetric efficiencies are derived from the reconstructed thermodynamic states.

The isentropic efficiency is calculated as:

$$\eta_{\text{is}} = \frac{h_{2s} - h_1}{h_2 - h_1} = \frac{h_{2s} - h_1}{P_c / \dot{m}}, \quad (24)$$

where h_{2s} is the compressor outlet enthalpy for an isentropic compression from P_{evap} to P_{cond} .

The volumetric efficiency is calculated as:

$$\eta_{\text{vol}} = \frac{\dot{m}}{\rho_1 \dot{V}_{\text{swept}}}, \quad (25)$$

where ρ_1 is the refrigerant density at the compressor inlet and $\dot{V}_{\text{swept}}$ is the compressor swept volume rate.

4 HEAT EXCHANGER MODEL AND NUMERICAL SOLUTION

4.1 LMTD HEAT EXCHANGER MODEL

The evaporator and condenser are modeled as counterflow heat exchangers. Pressure drops are neglected, and the refrigerant-side temperature is assumed approximately constant during phase change. The heat transfer rate is calculated using the LMTD method:

$$Q = KA\Delta T_{\text{lm}}, \quad (26)$$

where KA is the overall heat conductance of the heat exchanger and ΔT_{lm} is the logarithmic mean temperature difference:

$$\Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}. \quad (27)$$

For the evaporator, heat is transferred from the air stream to the refrigerant. The terminal temperature differences are defined as:

$$\Delta T_{1,\text{evap}} = T_{\text{air,in}} - T_{\text{evap}}, \quad (28)$$

$$\Delta T_{2,\text{evap}} = T_{\text{air,out}} - T_{\text{evap}}. \quad (29)$$

Therefore, the evaporator heat balance is:

$$Q_e = KA_{\text{evap}}\Delta T_{\text{lm,evap}}. \quad (30)$$

For the condenser, heat is rejected from the refrigerant to the water stream. The terminal temperature differences are:

$$\Delta T_{1,\text{cond}} = T_{\text{cond}} - T_{\text{w,out}}, \quad (31)$$

$$\Delta T_{2,\text{cond}} = T_{\text{cond}} - T_{\text{w,in}}. \quad (32)$$

Therefore, the condenser heat balance is:

$$Q_c = KA_{\text{cond}}\Delta T_{\text{lm,cond}}. \quad (33)$$

4.2 HEAT EXCHANGER CONDUCTANCES

The heat exchanger conductances are calculated at the nominal operating point and then kept constant throughout the parametric study. This follows the project assumption that the same evaporator and condenser are used while the heat-sink temperature varies.

At the nominal condition, the calculated conductance values are:

Table 7: Heat exchanger conductances used in the model

Parameter	Value
KA_{evap}	3.13 kW/K
KA_{cond}	12.61 kW/K

These values are obtained from:

$$KA_{\text{evap}} = \frac{Q_e}{\Delta T_{\text{lm, evap}}}, \quad (34)$$

$$KA_{\text{cond}} = \frac{Q_c}{\Delta T_{\text{lm, cond}}}. \quad (35)$$

4.3 PARAMETRIC OPERATING CONDITIONS

The parametric study is performed by varying the condenser water inlet temperature from 20°C to 40°C, with a step of 2°C. The water temperature rise across the condenser is fixed at 5 K:

$$T_{\text{w, out}} = T_{\text{w, in}} + 5 \text{ K}. \quad (36)$$

All other main assumptions, including superheating, subcooling, refrigerant, compressor model, and heat exchanger conductances, are kept fixed.

Table 8: Parametric study settings

Quantity	Value
Variable parameter	$T_{\text{w, in}}$
Range	20°C–40°C
Step	2°C
Water temperature rise	5 K
Evaporator air inlet/outlet	24°C/12°C
Superheating	6 K
Subcooling	3 K

4.4 ITERATIVE SOLUTION METHOD

For each value of condenser water inlet temperature, the evaporating and condensing temperatures must satisfy both the compressor model and the heat exchanger balances. Since the compressor performance depends on T_{evap} and T_{cond} , and the heat exchanger balances also determine these temperatures, the problem is solved iteratively.

The solution procedure is:

1. Guess initial values of T_{evap} and T_{cond} .
2. Evaluate the compressor polynomial to obtain Q_e , P_c , and $\dot{m}$.
3. Reconstruct the thermodynamic cycle using CoolProp.

4. Calculate Q_e and Q_c from the cycle energy balances.
5. Update T_{evap} and T_{cond} from the evaporator and condenser LMTD equations.
6. Repeat until the temperature changes between two consecutive iterations are below the convergence tolerance.

The convergence criterion is:

$$\max \left(\left| T_{\text{evap}}^{(k+1)} - T_{\text{evap}}^{(k)} \right|, \left| T_{\text{cond}}^{(k+1)} - T_{\text{cond}}^{(k)} \right| \right) < 10^{-4} \text{ K}. \quad (37)$$

A maximum of 100 iterations is imposed. In the performed simulations, convergence was reached within 10–14 iterations over the studied heat-sink temperature range.

5 RESULTS AND DISCUSSION

5.1 NOMINAL COMPRESSOR VALIDATION

The compressor polynomial model was first validated against the nominal Bitzer operating point used for compressor selection. The comparison is reported in Table 9. The close agreement between the manufacturer values, polynomial outputs, and thermodynamic reconstruction confirms that the polynomial coefficients were implemented correctly.

Table 9: Nominal compressor validation at $T_{\text{evap}} = 2^\circ\text{C}$ and $T_{\text{cond}} = 37.5^\circ\text{C}$

Quantity	Manufacturer	Polynomial	Thermodynamic reconstruction
Q_e [kW]	47.700	47.722	47.712
P_c [kW]	9.690	9.687	9.687
$\dot{m}$ [kg/h]	664.000	663.583	663.583
EER/COP [-]	4.920	4.926	4.925

The thermodynamic reconstruction gives $Q_e = 47.712$ kW, very close to the manufacturer value of 47.700 kW. The same consistency is obtained for compressor power, refrigerant mass flow rate, and EER. This validates the use of the polynomial model within the considered operating range.

5.2 REFERENCE CYCLE STATES

The reference case is taken at a condenser water inlet temperature of 30°C , corresponding to a condenser water outlet temperature of 35°C . The converged refrigerant-side temperatures are:

$$T_{\text{evap}} = 2.00^\circ\text{C}, \quad T_{\text{cond}} = 37.50^\circ\text{C}. \quad (38)$$

The corresponding thermodynamic states are shown in Table 10.

Table 10: Reference cycle states at $T_{\text{w,in}} = 30^\circ\text{C}$

State	T [$^\circ\text{C}$]	P [bar]	h [kJ/kg]	s [kJ/(kg K)]
1	8.00	8.665	523.009	2.1735
2	80.43	23.307	575.564	2.2064
3	34.50	23.307	264.165	1.2156
4	2.00	8.665	264.165	1.2331

State 1 is superheated vapor at the compressor inlet, while state 3 is subcooled liquid at the condenser outlet. The expansion process is isenthalpic, as shown by $h_4 = h_3$. The real compression process increases pressure, temperature, enthalpy, and entropy.

5.3 EFFECT OF HEAT-SINK TEMPERATURE

The condenser water inlet temperature was varied from 20°C to 40°C. A compact summary of representative results is reported in Table 11. The full numerical sweep was performed with a step of 2°C.

Table 11: Parametric results for selected condenser water inlet temperatures

$T_{w,in}$ [°C]	T_{evap} [°C]	T_{cond} [°C]	Q_e [kW]	P_c [kW]	EER [-]	PR [-]
20	1.18	27.56	50.41	7.86	6.42	2.14
24	1.50	31.54	49.36	8.55	5.78	2.35
28	1.83	35.51	48.27	9.29	5.20	2.57
30	2.00	37.50	47.71	9.69	4.93	2.69
34	2.35	41.48	46.56	10.53	4.42	2.93
40	2.90	47.45	44.72	11.94	3.74	3.32

Full results were calculated every 2°C from 20°C to 40°C. The table reports representative values for compactness.

As the condenser water inlet temperature increases, the condensing temperature also increases. This raises the condensing pressure and therefore the pressure ratio. Over the studied range, the pressure ratio increases from 2.14 to 3.32.

The higher pressure ratio increases the compressor work. As a result, the compressor power rises from 7.86 kW to 11.94 kW:

$$\Delta P_c = \frac{11.94 - 7.86}{7.86} \times 100 = 52.0\%. \quad (39)$$

At the same time, the cooling capacity decreases from 50.41 kW to 44.72 kW:

$$\Delta Q_e = \frac{44.72 - 50.41}{50.41} \times 100 = -11.3\%. \quad (40)$$

The combined effect of lower cooling capacity and higher compressor power causes a significant decrease in EER:

$$\Delta \text{EER} = \frac{3.74 - 6.42}{6.42} \times 100 = -41.6\%. \quad (41)$$

Therefore, the results confirm the expected behavior: increasing the heat-sink temperature worsens the refrigeration cycle performance by increasing compressor work and reducing the useful cooling effect.

5.4 COMPRESSOR EFFICIENCIES

The isentropic and volumetric efficiencies are not imposed as constants. They are derived from the polynomial compressor outputs and the reconstructed thermodynamic states. The values obtained over the operating range are summarized in Table 12.

Table 12: Derived compressor efficiencies for selected operating points

$T_{w,in}$ [°C]	PR [-]	η_{is} [-]	η_{vol} [-]	T_{cond} [°C]
20	2.14	0.717	0.980	27.56
24	2.35	0.750	0.977	31.54
28	2.57	0.774	0.973	35.51
30	2.69	0.782	0.971	37.50
34	2.93	0.793	0.968	41.48
40	3.32	0.796	0.961	47.45

The efficiencies are derived quantities, not fixed input assumptions.

The isentropic efficiency increases from approximately 0.717 to 0.796 over the studied range. This behavior comes from the manufacturer compressor map: the compressor operates at different combinations of evaporating and condensing temperatures, and the effective isentropic performance changes accordingly.

The volumetric efficiency decreases slightly from approximately 0.980 to 0.961. This is physically reasonable because a higher pressure ratio generally increases re-expansion and internal leakage effects, reducing the effective intake of refrigerant vapor per swept volume.

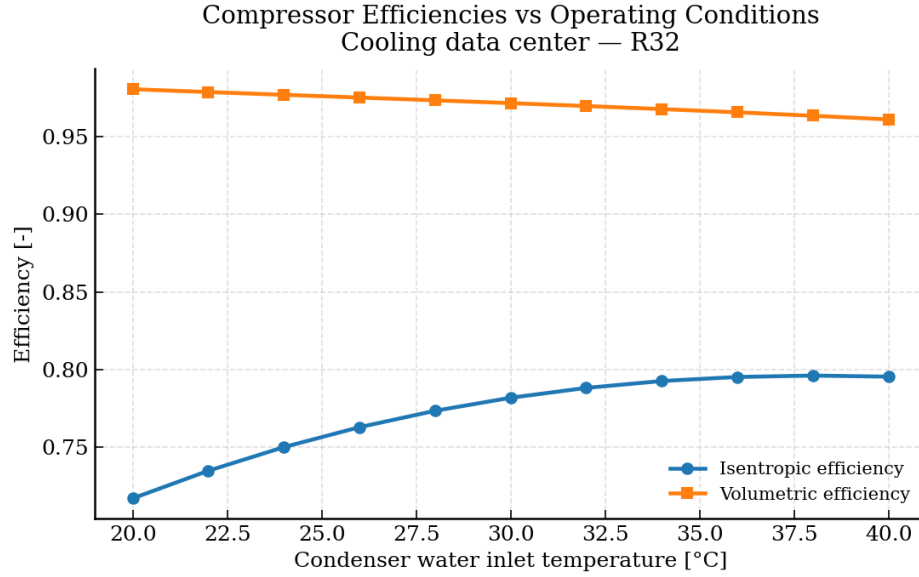


Figure 3: Derived isentropic and volumetric efficiencies as functions of condenser water inlet temperature.

The efficiency trends confirm that the compressor behavior is not modeled with fixed efficiencies. Instead, the operating-condition dependence is captured through the Bitzer polynomial map and the CoolProp-based thermodynamic reconstruction.

5.5 SECONDARY-FLUID MASS FLOW RATES

The air and water mass flow rates are calculated from the heat duties and imposed secondary-fluid temperature differences:

$$\dot{m}_{\text{sec}} = \frac{Q}{c_p \Delta T}. \quad (42)$$

At the reference condition $T_{w,\text{in}} = 30^\circ\text{C}$, the evaporator air mass flow rate and condenser water mass flow rate are:

$$\dot{m}_{\text{air}} = 3.96 \text{ kg/s}, \quad \dot{m}_{\text{water}} = 2.75 \text{ kg/s}. \quad (43)$$

These values correspond to the imposed air-side temperature decrease from 24°C to 12°C and the imposed condenser water temperature rise of 5 K.

5.6 THERMODYNAMIC DIAGRAMS

The pressure–enthalpy and temperature–entropy diagrams are used to visualize the cycle at selected operating conditions. The diagrams show the main vapor-compression processes: compression, condensation, expansion, and evaporation.

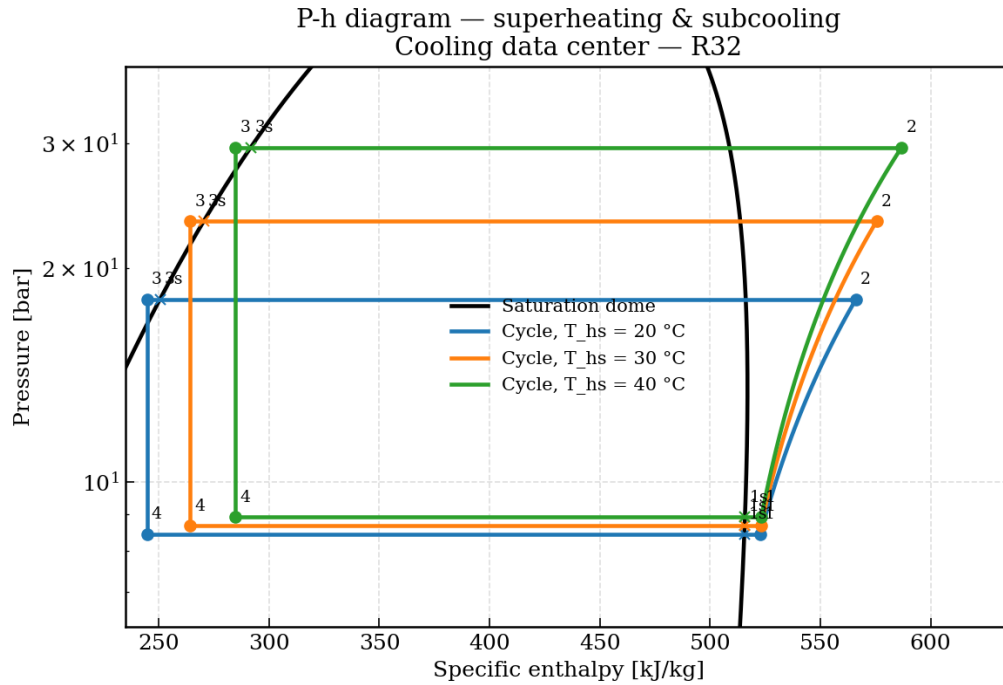


Figure 4: Pressure–enthalpy diagram for selected operating conditions.

In the P–h diagram, higher condenser water temperature shifts the high-pressure side of the cycle upward. This reflects the increase in condensing pressure and compressor work.

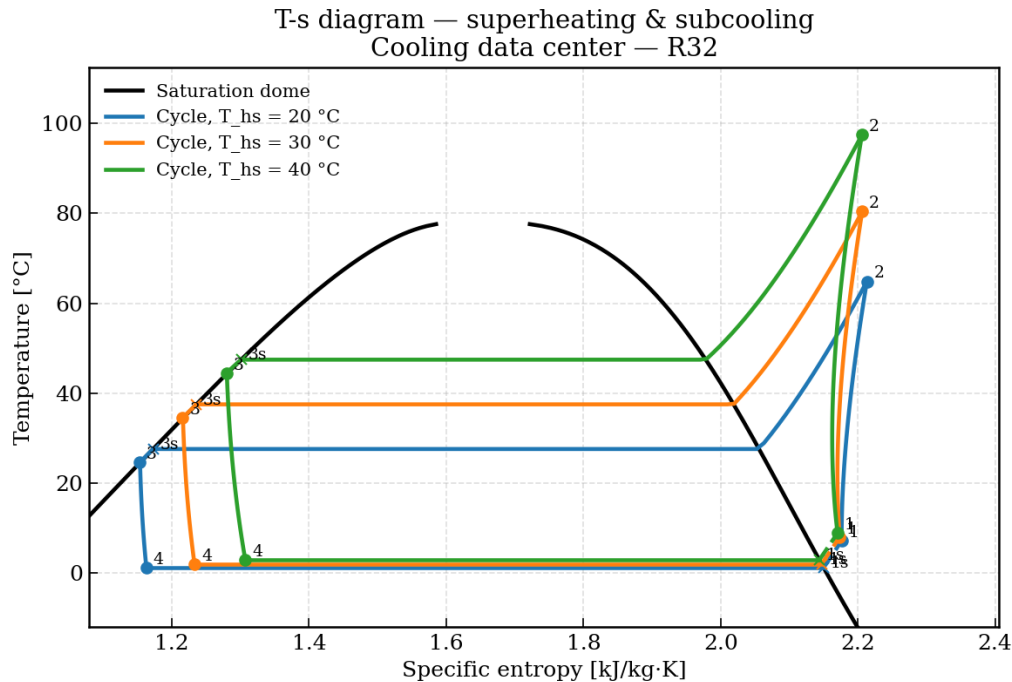


Figure 5: Temperature–entropy diagram for selected operating conditions.

The T–s diagram highlights the imposed superheating at the compressor inlet, subcooling at the condenser outlet, and entropy increase during the real compression process.

5.7 PERFORMANCE PLOTS

The main performance trends are shown in Figures 6–9. These plots summarize the effect of condenser water inlet temperature on system performance.

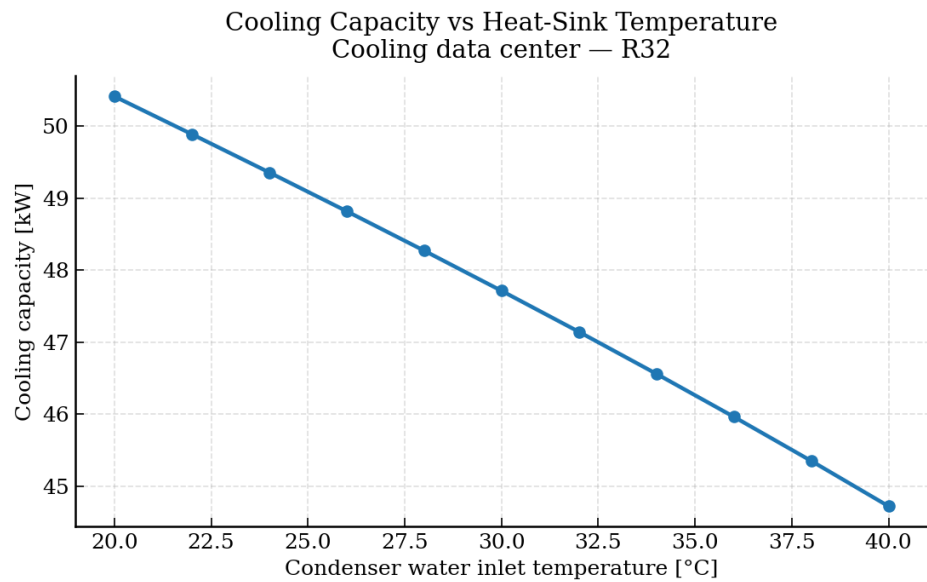


Figure 6: Cooling capacity as a function of condenser water inlet temperature.

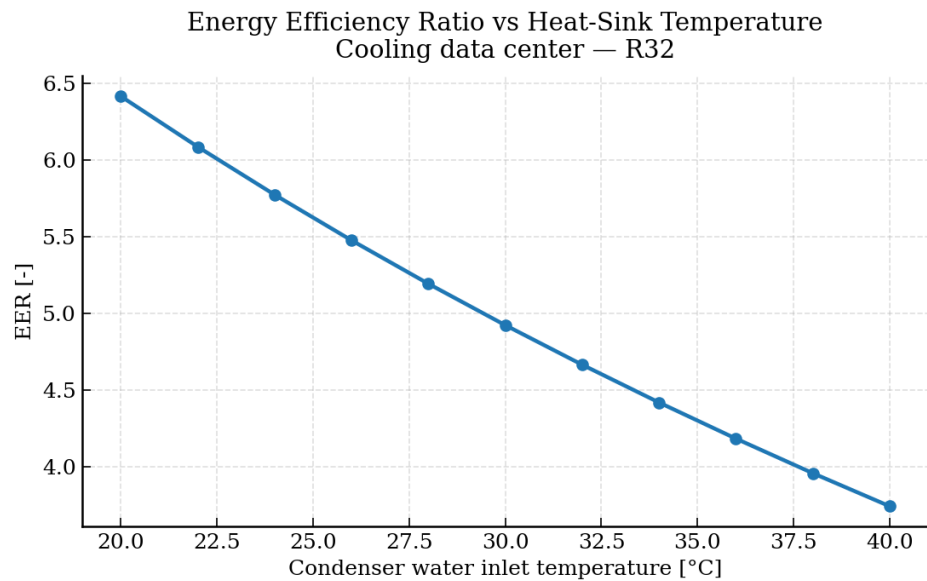


Figure 7: Energy Efficiency Ratio as a function of condenser water inlet temperature.

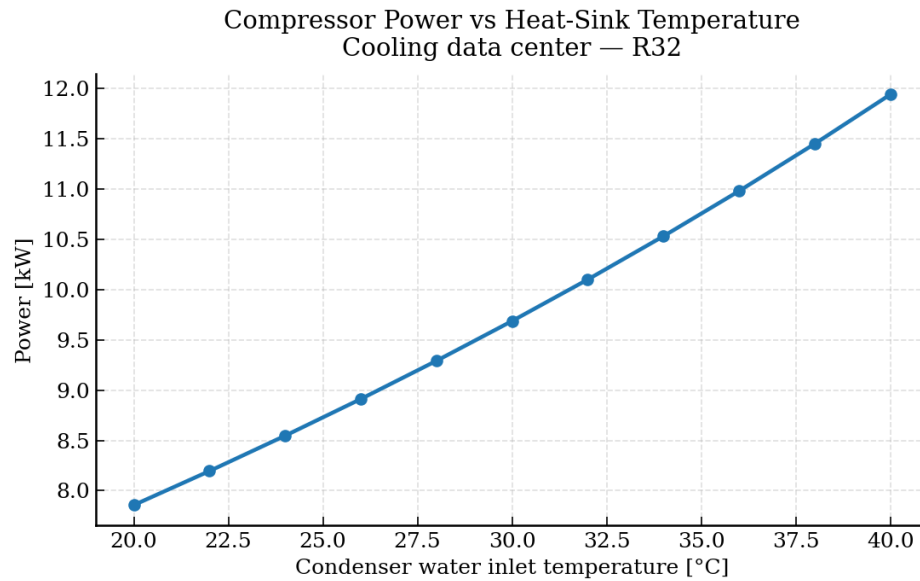


Figure 8: Compressor power as a function of condenser water inlet temperature.

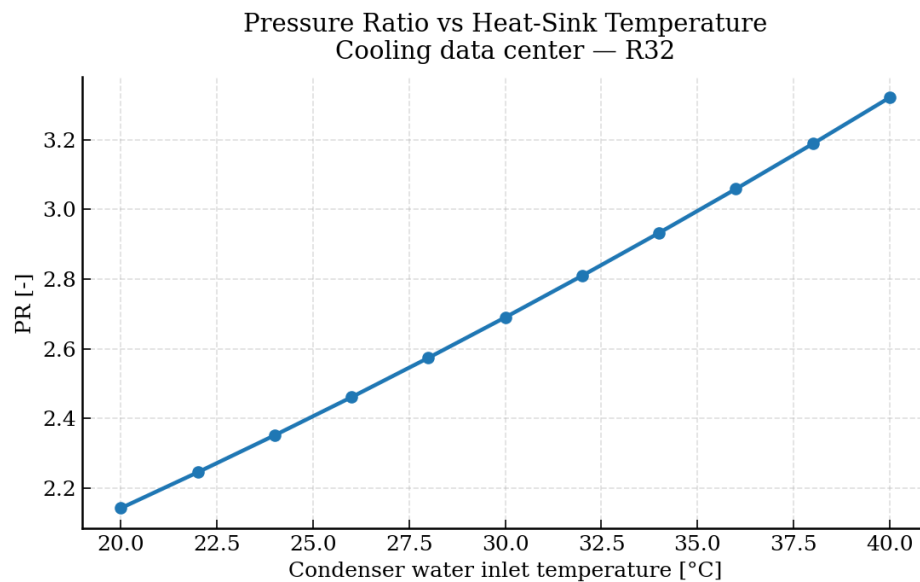


Figure 9: Pressure ratio as a function of condenser water inlet temperature.

The plots show that a colder heat sink improves system performance, while a warmer heat sink increases pressure ratio and compressor power and reduces both cooling capacity and EER.

6 CONCLUSION

A vapor-compression refrigeration system for data center cooling was modeled and analyzed. The assigned nominal cooling capacity was 50 kW, with air as the evaporator secondary fluid and water as the condenser secondary fluid.

A Bitzer GSU60182VL_4 scroll compressor was selected using R32 as the refrigerant. At the nominal compressor-selection point, the selected compressor provides 47.7 kW, corresponding to a deviation of -4.6% from the assigned

capacity. This was considered acceptable because it was the closest available Bitzer R32 scroll option to the 50 kW target, whereas the next larger option would have provided 60.2 kW, corresponding to an oversizing of +20.4%.

The compressor polynomial model was validated against the Bitzer nominal data. The thermodynamic reconstruction gave $Q_e = 47.712$ kW, $P_c = 9.687$ kW, and $EER = 4.925$, showing very close agreement with the manufacturer data.

The parametric study showed that increasing the condenser water inlet temperature from 20°C to 40°C significantly worsens the system performance. Over this range:

- cooling capacity decreased from 50.41 kW to 44.72 kW, corresponding to −11.3%;
- compressor power increased from 7.86 kW to 11.94 kW, corresponding to +52.0%;
- EER decreased from 6.42 to 3.74, corresponding to −41.6%;
- pressure ratio increased from 2.14 to 3.32.

These results confirm the expected behavior of the refrigeration cycle: a warmer heat sink increases the condensing temperature and pressure ratio, which increases compressor work and reduces the overall efficiency. Therefore, condenser-side operating conditions have a strong influence on the cooling capacity and EER of the system.

APPENDIX

A. POLYNOMIAL COEFFICIENTS

The compressor performance was represented using the 10-coefficient polynomial exported from the Bitzer selection software for the selected Bitzer GSU60182VL.4 compressor with R32.

$$y = c_1 + c_2 T_{\text{evap}} + c_3 T_{\text{cond}} + c_4 T_{\text{evap}}^2 + c_5 T_{\text{evap}} T_{\text{cond}} + c_6 T_{\text{cond}}^2 + c_7 T_{\text{evap}}^3 + c_8 T_{\text{cond}} T_{\text{evap}}^2 + c_9 T_{\text{evap}} T_{\text{cond}}^2 + c_{10} T_{\text{cond}}^3 \quad (44)$$

The polynomial is used independently for cooling capacity Q_e , compressor power P_c , and refrigerant mass flow rate $\dot{m}$. Temperatures are expressed in °C. The mass flow rate is returned in kg/h and converted to kg/s in the model.

Table 13: Polynomial coefficients for Bitzer GSU60182VL.4

Output	Coefficients ($c_1, \dots, c_{10}$)
Q_e [W]	59389.6461, 1974.9334, -434.5662, 24.5945, -10.2182, 2.2480, 0.1388, -0.1284, -0.0268, -0.0326
P_c [W]	3944.2859, 31.3520, 132.7176, 2.6430, -0.1076, -0.3117, 0.0556, -0.0364, -0.0014, 0.0218
$\dot{m}$ [kg/h]	658.9665, 21.4502, -1.9168, 0.2629, -0.0208, 0.0397, 0.0021, 0.0002, 0.0003, -0.0004

The polynomial variables are T_{evap} and T_{cond} in °C. The mass flow rate is returned in kg/h and converted to kg/s in the model.

B. CODE REPOSITORY AND IMPLEMENTATION

The numerical model was implemented in Python and is available in the following GitHub repository:

<https://github.com/Amirhossein-Naeejnezhad/refrigeration-cycle-model>

The final version used for this report is contained in the branch:

`universal-model`

The code reproduces the complete workflow of the project: compressor polynomial evaluation, CoolProp thermodynamic reconstruction, LMTD heat exchanger matching, iterative solution of the operating point, parametric analysis, and generation of the tables and figures used in the report.

Table 14: Summary of the code structure

File group	Purpose
<code>src/config.py</code>	Project inputs: refrigerant, compressor, temperatures, assumptions
<code>src/compressor.py</code>	Bitzer polynomial coefficients and compressor-map evaluation
<code>src/thermodynamics.py</code>	CoolProp state calculation and cycle reconstruction
<code>src/heat_exchanger.py</code> , <code>src/solver.py</code>	LMTD heat exchangers and iterative coupling
<code>src/plots*.py</code>	Performance plots, P-h diagram, and T-s diagram
<code>src/main.py</code> , <code>run_project.ipynb</code>	Main execution workflow and exported result tables

The notebook exports the main numerical results as CSV files, including the parametric study, nominal validation table, and reference thermodynamic states. The implementation is modular: changing the project case only requires editing the configuration file and compressor coefficient data, while the thermodynamic, heat exchanger, and solver logic remain unchanged.

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